Exp 10: Write a Servlet program to store a users preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits. DATE:

AIM:

To create a Java Servlet that stores user preferences such as theme and language using cookies, and retrieves and displays these preferences when the user revisits the application.

ALGORITHM:

1. Input:

- The user selects preferences such as:
  - Theme (Light/Dark)
  - Language (English/Tamil)
- The form submits this data to the servlet.

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2. Steps for Creating and Handling Cookies in Servlet:

Step 1: Create an HTML Form

- Design an HTML form with:
  - Dropdown for selecting theme
  - Dropdown for selecting language
- Add a submit button
- Set the action attribute to the servlet URL
- Use the POST method

Step 2: Create the Servlet to Handle Preferences

- Write a Java Servlet class
- Override the following methods:

- doPost() – to store preferences
- doGet() – to retrieve preferences

Step 3: Store Preferences Using Cookies (doPost method)

- Retrieve user inputs using:

```
request.getParameter("theme");
request.getParameter("language");
```

- Create cookies:

```
Cookie themeCookie = new Cookie("theme", theme);
Cookie langCookie = new Cookie("language", language);
```

- Set cookie expiry time:

```
themeCookie.setMaxAge(60 * 60 * 24);
langCookie.setMaxAge(60 * 60 * 24);
```

- Add cookies to response:

```
response.addCookie(themeCookie);
response.addCookie(langCookie);
```

Step 4: Retrieve Preferences Using Cookies (doGet method)

- Get cookies from request:

```
Cookie[] cookies = request.getCookies();
```

- Loop through cookies and extract values:

```
for (Cookie c : cookies) {
   if (c.getName().equals("theme")) {
      theme = c.getValue();
   }
   if (c.getName().equals("language")) {
      language = c.getValue();
   }
}
```

Step 5: Display Preferences

- Generate HTML response showing:
    - Selected theme
    - Selected language
- If cookies exist → display saved preferences
- If cookies do not exist → display default message

Step 6: Deploy and Test the Servlet

- Compile the servlet
- Deploy in a server (e.g., Apache Tomcat)
- Open browser and submit preferences
- Refresh or revisit to check stored values

Step 7: Observe Results

- First visit → preferences are stored in cookies
- Subsequent visits → preferences are retrieved automatically

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PROGRAM:

HTML Code (index.html)

```
<!DOCTYPE html>
<html>
<head>
  <title>User Preferences</title>
</head>
<body>

<h2>Select Preferences</h2>

<form action="PreferenceServlet" method="post">

  Theme:
  <select name="theme">
    <option value="Light">Light</option>
```

```
        <option value="Dark">Dark</option>
    </select>
    <br><br>

    Language:
    <select name="language">
        <option value="English">English</option>
        <option value="Tamil">Tamil</option>
    </select>
    <br><br>

    <input type="submit" value="Save Preferences">

</form>

</body>
</html>
```

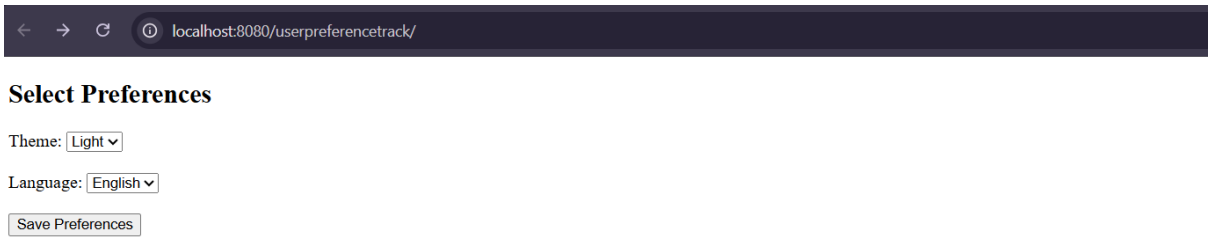
Servlet Code (PreferenceServlet.java)

```
package com.example;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/PreferenceServlet")
public class PreferenceServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String theme = request.getParameter("theme");
        String language = request.getParameter("language");

        Cookie themeCookie = new Cookie("theme", theme);
        Cookie langCookie = new Cookie("language", language);
```

```
        themeCookie.setMaxAge(60 * 60 * 24);
        langCookie.setMaxAge(60 * 60 * 24);

        response.addCookie(themeCookie);
        response.addCookie(langCookie);

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2>Preferences Saved!</h2>");
        out.println("<a href='" + request.getContextPath() + "/PreferenceServlet'>View
Preferences</a>");
    }

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String theme = "Not set";
        String language = "Not set";

        Cookie[] cookies = request.getCookies();

        if (cookies != null) {
            for (Cookie c : cookies) {
                if (c.getName().equals("theme")) {
                    theme = c.getValue();
                }
                if (c.getName().equals("language")) {
                    language = c.getValue();
                }
            }
        }

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2>User Preferences</h2>");
        out.println("Theme: " + theme + "<br>");
        out.println("Language: " + language);
    }
}
```

OUTPUT:

- The user selects theme and language from the form.
- Preferences are stored using cookies.
- On revisiting, stored preferences are displayed.



localhost:8080/userpreferencetrack/PreferenceServlet

Preferences Saved!

View Preferences

localhost:8080/userpreferencetrack/PreferenceServlet

User Preferences

Theme: Light
Language: English

RESULT:

A Servlet program is successfully created to store user preferences using cookies and retrieve them on subsequent visits.